H.T.No. | | | | | | | | | | |

Code No: CT4523 **SRGEC-R23**

III B.Tech I Semester Regular Examinations, November 2025

DATA WAREHOUSING AND DATA MINING

(Computer Science and Engineering & Artificial Intelligence and Data Science)

Time: 3 Hours **Max. Marks: 70**

Note: All questions from **PART-A** are to be answered at one place.
Answer one question from each unit of **PART-B.** All questions carry equal marks.

PART-A

10 × 2 = 20M

1. a) Define snowflake schema
   b) State the role of Concept Hierarchies in OLAP.
   c) State entity identification problem.
   d) Differentiate between missing values and noisy data.
   e) State Bayes theorem.
   f) What is the general approach to solving a classification problem?
   g) What is compact representation of frequent itemsets?
   h) Give few techniques to improve the efficiency of Apriori algorithm.
   i) What is the purpose of the K-means algorithm?
   j) Calculate the distance between two objects, A = (10, 4, 2, 10) and B = (4, 2, 1, 4) using the Manhattan and Euclidean distance measures.

PART-B

5 × 10 = 50M

UNIT-I

2. Analyze how visualization techniques enhance Data Analysis in OLAP environments. (10M)

(OR)

3. a) Illustrate the OLAP operations in Data Cube. (6M)
   b) Explain how Cloud Data Warehousing supports scalability. (4M)

UNIT-II

4. a) Describe attribute subset selection and its role in Data Preprocessing. (5M)
   b) Explain the concept of histograms for data reduction. (5M)

(OR)

5. a) Describe entity identification problems in data integration. (5M)
   b) Explain conflict detection and resolution in integrated datasets. (5M)

UNIT-III

6. a) Discuss the process of rule extraction from decision trees. (6M)
   b) Explain Bayes Therorem with example. (4M)

(OR)

7. a) Discuss the importance of evaluation metrics such as Accuracy, Precision, Recall, and F1-score. (6M)
   b) Explain the Decission Tree Induction algorithm. (4M)

UNIT-IV

8. Given the following transaction data, find the frequent itemsets usingApriori algorithm if minimum support is 30% and confidence is 60%. (10M)

TID	Items
1	{Jacket, Boots}
2	{Milk, Cheese, Bread, Shoes}
3	{Cloth, Bread}
4	{Milk, Bread, Shoes, Pork, Apple, Soup}
5	{Cheese, Shoes, Beef}
6	{Jacket, Bread, SkiPants}

(OR)

9. Given the following database with 5 transactions and a minimum support threshold of 60% and a minimum confidence threshold of 80%, compute all frequent itemsets using Apriori algorithm by generating candidate itemsets. List all strong association rules obtained from frequent itemsets. Specify, how FP-Growth is better than Apriori? (10M)

Trans Id	Itemset
1	M,O,N,K,E,Y
2	D,O,N,K,E,Y
3	M,A,K,E
4	M,U,C,K,Y
5	C,O,L,I,E

UNIT-V

10. Suppose that the data mining task is to cluster the following eight points (with (x, y) representing location) into three clusters: (10M)

    A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Manhattan distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show only

    (i) The three cluster centers after the first-round execution. (ii) The final three clust

(OR)

11. For the given dataset which represents 2D points in a plane:

Point	X	Y
A	1	2
B	2	2
C	2	3
D	8	7
E	8	8
F	25	80

    Apply the DBSCAN algorithm to this dataset using eps (ε) = 2, and min_samples = 2.

    Determine: (i) The core points, border points, and noise points. (ii) The final clusters formed.

    (iii) Plot the points and visually show the clusters (10M)

*****